

Project Title	Underwater Lidar Turbidity System	Start Date	8/27/2026
Project Lead	Dylan Tappendorf	Prioritization	Developing a working Lidar prototype capable of working Underwater
Other Owners	Alexander Strattard, Alexander Paquette	Projected finish date	12/12/2026

Purpose: The purpose of this project is to create a Lidar system capable of operating underwater in order to take measurements of water turbidity in real time using cheap and off the shelf components. The project has real world and business applications because the only other options for such a machine cost somewhere in the ballpark of 80,000 dollars.

Objectives: Success Criteria for this project are the production of a working prototype that is able to accurately use the phase difference obtained by the LiDAR in order to calculate the turbidity of water.

Constraints: This project has a variety of important constraints that limit its development such as a 15 week deadline for total development. The development of the system also has to obey certain cost constraints as the whole crux of the project rests on the price being cheaper than current alternatives.

Time/Decision Points: The total time allotted to the project is 15 weeks which are broken up into a variety of milestones in which progress is required to be submitted. Most of the decisions related to design are expected to be made early on and will be tested in the testing phase. Decisions made related to design are to be made via consensus of all involved members.

Measurable success criteria: The design will be considered successful if the accuracy of the water turbidity is at least 80% but ideally higher than that while also being significantly cheaper than current alternatives ideally not costing more than about \$500 per unit.